1. Calculate the following indefinite integrals.

a) $\int x\sqrt{1+x^2}\,dx$

b) $\int \frac{\sin\left(\sqrt{x}\right)}{2\sqrt{x}}\,dx$

c) $\int \frac{2+\ln(x)}{x}\,dx$

d) $\int \frac{2x}{1+x^4}\,dx$

e) $\int x^2\left(1-x^3\right)^{10}\,dx$

2. Suppose that $f$ is an odd function and $g$ is an even function which are each integrable on the interval $[-3,3]$. Given that $\int_0^3 f(x)\,dx = 4$ and $\int_0^3 g(x)\,dx = 5$, evaluate the following definite integral.

$\int_{-3}^{3}\left[6f(x)+8g(x)\right]dx$

3. Calculate the following definite integrals.

a) $\int_0^{\pi/2} \sin(x)\cos(x)\sqrt{1-\sin(x)}\,dx$

b) $\int_0^1 x^3\sqrt{3+x^2}\,dx$

c) $\int_0^{\pi/4} \cot(x)\,dx$

d) $\int_{\pi/4}^{\pi/2} \cot(x)\, dx$

e) $\int_{0}^{1} x^2 \sqrt{1-x}\, dx$

f) $\int_{-e}^{e} \frac{x^3}{\ln\left(x^2+2\right)}\, dx$